Envisioner Project — Handoff (Stair System Complete)

Project

Envisioner — a floor-plan/room design tool with 2D canvas (Canvas2D.jsx) and 3D view (View3D.jsx), built with React + Three.js. Supports walls, doors, windows, furniture, roofs/cantilevers, multi-floor designs, and now a full multi-type stair system.

Status: Stair system is DONE and confirmed working

The multi-type stair system (Straight, L-shaped, U-shaped) has been fully implemented and verified after multiple rounds of fixes. Confirmed working:

- Stair placement with topmost-floor prevention ("Add another floor above this floor before placing stairs.")
- Full stair schema (id, type, fromFloorId, toFloorId, x, y, rotation, width, run, landingLength, turnDirection, stepCount, color)
- Select, move, free-rotate (drag handle), R-key 15° rotation, and delete for stairs — same interaction pattern as furniture
- Immediate red collision warnings in both 2D and 3D (stair vs wall/door/window/furniture/other stairs), no delay, no need to select first
- Accurate multi-part L/U footprint polygons for collision + rendering (not simplified bounding boxes)
- Room-boundary validation (stair must fit inside a closed room)
- Upper-floor dashed, non-editable "Stairs from [floor name]" arrival guide
- Procedural ceiling/floor-slab opening above stairs, shaped to match the real stair footprint (fixed a bug where L/U holes rendered as a triangle due to multiple overlapping THREE.Shape holes — fixed by merging edge-sharing rectangles into one unified outline polygon per stair before passing to THREE.Shape)
- Single-gesture undo/redo (one history entry per completed move/rotate/delete/placement, never per-frame)
- Visible/selectable in Normal, X-Ray, Hide Walls, and All Floors view modes
- 3D groups tagged with `userData.kind = "staircase"` and `userData.stairId` for future GLB export
- No structural/roof/cantilever modification from the ceiling-opening feature — it's render-only

Bugs fixed along the way (for reference if anything regresses)

1. **Blue screen bug** — occurred first via stairs, then regressed via furniture placement. Root cause: `overlappingIds` referenced in Canvas2D.jsx but the prop was actually named `overlappingFurnIds` — calling `.has()` on `undefined` threw a TypeError that crashed the React render, exposing the dark canvas background. Fixed by correcting the reference.
2. **Stairs frozen after placement** (unselectable/unmovable/undeletable) — was a side effect of the same blue-screen crash; fixing the `overlappingIds` reference resolved this too.
3. **Wall endpoint resize broken** — Canvas2D destructured prop names (`onEndpointMoveStart`, etc.) didn't match the names actually referenced in handler code (`onEndpointStart`, etc.), causing a silent ReferenceError. Fixed by reconciling naming.
4. **Ceiling hole for L/U stairs rendered as a triangle** — caused by passing 3 separate edge-sharing rectangles as 3 separate THREE.Shape holes, which broke earcut triangulation. Fixed by merging them into one outline polygon first.
5. **Furniture collision with stairs was one-sided** (stair showed red, furniture didn't) — fixed by centralizing validation into a shared `computeFloorValidations` function so both entities update symmetrically.
6. **False "unsupported wall" warning on corner-to-corner upper-floor walls** — classic ray-cast point-in-polygon rejected boundary-touching points. Fixed with an epsilon-tolerant point-on-segment check before ray casting.
7. **R key didn't rotate stairs**, only furniture — added parallel handler.

Known environment issue (unrelated to app code)

Antigravity's built-in browser-testing tool (bundled playwright-go, hardcoded driver v1.57.0) fails with a 404 because Microsoft retired the Azure CDN it downloads from. This is a confirmed, still-open Antigravity bug (tracked upstream), not an Envisioner bug. Workaround: manually place a matching playwright-core driver from npm at the expected local cache path, or install a standalone `@playwright/mcp` server as an MCP connector instead of relying on the built-in browser tool. Worth re-applying this workaround in the new thread if browser-based verification is needed again.

Bugs reported but not yet confirmed fixed (carry into new thread if still open)

As of the last check-in, these three were reported and a fix plan was sent, but final confirmation of their resolution was not yet received in this thread:

- **Bug E**: 2D and 3D furniture rotation disagree — sidebar showed 321° in one screenshot and 122° in another for what looked like related states; likely a sign/axis mismatch between how Canvas2D applies rotation (SVG transform) and how View3D applies it (Three.js mesh rotation).
- **Bug F**: Wall drawn on an upper floor, extended clearly outside the lower floor's footprint, still showed a false "unsupported" warning — needs verification of actual sample-point coordinates to confirm if this is a genuine warning or a leftover bug from the boundary-epsilon fix.

- **Bug G**: Ceiling/roof on a cantilevered/extended wall section appeared visibly lower (not level) compared to the rest of the ceiling — likely a base-elevation or offset mismatch in cantilever roof height calculation.

If these were resolved after this note was written, disregard; otherwise, re-raise in the new thread with the same screenshots/description.

Working preferences learned this session

- User wants concrete root-cause confirmation (file/line references + "how did you verify it — code reading vs. actually run in browser") before accepting any bug is fixed. Multiple early rounds failed because fixes were declared based on code reading alone and turned out wrong.
- Prefer Claude Opus 4.5 for debugging/regression work in this codebase — it produced much more reliable root-cause analysis than Gemini 3.1 Pro High, which caused several regressions (frozen stairs, wall resize breaking, blue screen recurring) through narrow, poorly-scoped patches.
- Always require re-verification of previously-fixed behavior after any new change, since regressions have repeatedly slipped through in this project (stair interaction, wall resize, blue screen have each broken more than once).
- Do not push, commit, or deploy to GitHub unless explicitly requested.

Suggested opening message for the new thread

"Continuing work on Envisioner. The multi-type stair system (Straight/L-shaped/U-shaped) is complete and confirmed working after several rounds of bug fixes — see attached handoff notes for full history. [If bugs E/F/G are still open: paste their descriptions here.] What's next: [describe next feature/bug]."